

# Brandon Khumalo

## SOFTWARE DEVELOPER

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[Portfolio](#)

### PROFESSIONAL SUMMARY

Backend-focused fullstack developer with hands on experience designing and building scalable API's, systems, and fullstack applications using Python Django, Golang Echo, Next.js, React and Typescript. I design system architecture from backend API design and database modelling to frontend implementation and AWS cloud deployment. I have delivered production applications for real clients, including full e-commerce platforms with live payment integration and bring a strong foundation in clean architecture, maintainable codebases, and CI/CD with github Actions. I am passionate about solving real-world problems with technology and continuously growing as an engineer.

### SKILLS

**Programming Languages:** Java, Python, JavaScript, TypeScript, Golang

**Frameworks and Technologies:** Django Rest Framework, Echo (Go), Next.js, React, AWS, PostgreSQL

**Development Tools:** Git & Github, Android Studio, VSCode

**Soft Skills:** Adaptability, Critical Thinking, Problem-Solving, Time Management, Collaboration, Effective Communication

### Experience

Tishanyq Digital

2025 - current

#### ***Backend focused full-stack developer***

- Optimized application performance, scalability, and resource usage
- Ensured compliance with regulatory standards and industry best practices
- Ensured the systems are secure, scalable and reliable
- Built robust, scalable and high-performance applications
- Ensured application security and data integrity across all platforms
- Conducted code reviews, ensured adherence to industry standards
- Diagnosed and implemented solutions that balance speed, reliability and maintainability
- Deployed, implemented CI/CD pipelines and maintained applications on AWS
- Design system architecture and built robust, scalable, and high-performance applications with Python Django, Golang Echo, Next.js, TypeScript, React, and PostgreSQL

## Projects

### Laptop World — Full-Stack E-Commerce Platform

Client Project | March 2025 – April 2025 | [laptopworld.co.zw](https://laptopworld.co.zw)

- Built and deployed a production e-commerce platform end-to-end for a real Zimbabwean tech retailer using Next.js (App Router) and Django REST Framework with a decoupled frontend/backend architecture
- Implemented session-based cart sync via a UUID session key in a custom request header — no login required for shopping
- Integrated PayNow Zimbabwe for real payment processing, including async polling to confirm payment after order creation
- Engineered automatic image compression on every model save() across Category, Product, and ProductImage — handling a performance concern at the data layer
- Designed a 7-stage order lifecycle: awaiting\_payment → paid → processing → out\_for\_delivery → delivered/collected → cancelled
- Integrated transactional email via the Resend API for order confirmation workflows
- Built a Google Maps delivery address picker with GPS coordinates stored on the Order model and automatic delivery fee calculation based on location

**Tech Stack:** Next.js, TypeScript, Django REST Framework, PostgreSQL, TanStack Query, React Context, Zod, react-hook-form, Tailwind CSS, shadcn/ui, Framer Motion, Docker, PayNow, Google Maps API, Resend

### Online Voting System – Backend

*School Project | July 2025 – August 2025*

- Designed and implemented a scalable RESTFUL API backend for an online voting system using Django Rest Framework and PostgreSQL, enabling seamless poll creation, voting and real-time result computation.
- Architected optimized database schemas with appropriate indexing to support high-frequency vote operations and efficient data retrieval.
- Developed secure API endpoints with robust validation, preventing duplicate voting and restricting interactions on expired polls.
- Integrated real-time vote result broadcasting using Django Channels and WebSockets, providing live updates to frontend clients and enhancing user engagement..

- Authenticated and authorized API access through DRF permissions and token-based mechanisms to safeguard user actions.
- Wrote extensive unit and integration tests covering models, serializers, views, and routing to ensure reliability and maintainability of backend services.
- Containerized the application using Docker and Docker Compose, managing environment variables securely for development and production deployments.

**Tech Stack:** Python, Django, Django REST Framework, PostgreSQL, Redis, Django Channels, WebSockets, Docker, Swagger/OpenAPI, pytest, Git

## **TumaGo – Fullstack Mobile Package Delivery App**

### ***Personal Project | April 2025 – June 2025***

- Architected a hybrid microservice backend: Go API gateway, Go location and matching services for high-concurrency WebSockets and low-latency routing, and Django REST for business logic, admin, and payment workflows
- Built real-time driver tracking over WebSockets with Redis GEO indexing and PostgreSQL persistence, supporting [X] concurrent connections in load tests
- Implemented a driver matching engine using Dramatiq async workers and Redis GEO proximity search to surface the nearest available driver by vehicle type
- Developed two native Android applications (client and driver) in Java with booking flows, live GPS tracking, push notifications, earnings dashboards, and Google Maps/Places integration
- Implemented JWT-based auth across all services, per-IP rate limiting, internal endpoint protection, and graceful error fallbacks
- Built a FastAPI notification service and integrated Firebase Cloud Messaging for real-time push notifications on delivery status and driver messages

**Backend:** Go, Echo, Python, Django REST Framework, PgBouncer, Dramatiq, Redis, MySQL, JWT, Firebase Admin, AWS

**Frontend:** Java (Android), Retrofit, OkHttp, Firebase Messaging, Google Maps SDK, Places API, WebSockets

**DevOps & Tools:** Docker, Github Actions, Swagger

## **EDUCATION**

**ALX Africa**

**Jan 2025 - Aug 2025**

Certificate in Pro Backend Development

## **CERTIFICATES**

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|-----------------------------------------|-----------------------|
| ❖ AWS Cloud Practitioner                | <b>June 2024</b>      |
| ❖ Udemy Java Android Mobile Development | <b>March 2025</b>     |
| ❖ ALX Professional Foundations          | <b>April 2025</b>     |
| ❖ Google IT Support                     | <b>September 2025</b> |